

Design and Implementation of Real Time LPR System on a Fixed Point DSP

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Abstract—This paper presents a smart and simple algorithm for vehicle's license plate recognition system. The proposed algorithm consists of three major parts namely detection of license plate from an image, segmentation and recognition of characters. For detection of the license plate, a memory and speed proficient algorithm has been proposed. After detection, statistical based template matching is used for recognition of characters on the plate. The performance of the proposed algorithm is tested on real images. Experimental results illustrate that the proposed algorithm demonstrates enhanced performance in car license plate recognition.

Index Terms—License Plate Recognition, Line extraction, segmentation, optical character recognition

I. INTRODUCTION

License plate recognition (LPR) is a form of automatic vehicle identification. It uses image processing to identify vehicles by their license plates only.

An LPR system can be divided into three different parts (a) Extraction of the License Plate region from the input image (b) segmentations of characters and (c) recognition of segmented characters. A new technique is proposed for implementation of real time LPR system on a fixed point Digital Signal Processor.

There has been considerable interest in detection of the license plate including texture based extraction [9], color based extractions of license plate [1], boundary-line based extraction [2] and Hough transform for line detection[8]. Color based extraction fails if car comes in the same color as the license plate. On the other hand, Hough transform based approaches for boundary detection of the license requires a lot of resources (in terms of memory and time). In this paper, the boundary-line extraction methodology is chosen for license plate detection because of lesser memory requirement and deterministic plate boundary extraction.

After detection, the second step is to segment the license plate into sub-segments. Text region segmentation [4], [5] has been studied extensively during the past years but even today it remains an open field of work, interesting for different applications in which complex images are to be processed. Segmentation is usually done in two ways i.e. morphology based and connectivity based. Environmental effects like illumination changes can affect the license plate image which makes morphology based segmentation hard. On the other hand, Connectivity based segmentation fails where segments are disconnected. Therefore, a non conventional segmentation

approach is used which is known as Row and Column Sum approach.

The final step of proposed solution is to recognize the segmented characters. For this step usually neural network based Optical Character Recognition (OCR) [10] [3] is used [6], [11], [12] but it is costlier in terms of time consumption. Therefore, a Statistical based approach has been opted for character recognition. In this approach, decisions are taken on various features of detected characters. In this approach, compounds and end-points as features are used [7].

The balance of the paper is organized as follows: Section II describes the detection of license plate region, which is followed by segmentation of the license plate characters and digits in Section III. Procedure for the recognition of individual characters and digits is given in Section IV. Experimental results are presented in Section V. Finally, conclusion of the paper and guidelines for future work are provided in Section VI.

II. DETECTION OF LICENSE PLATE REGION

The proposed algorithm used for detection of licence plate region is based on the Boundary Line Extraction and the outline is shown in Fig. 1. The detection algorithm is used to detect the rectangular piece, which is the license plate, in the rear image of the car.

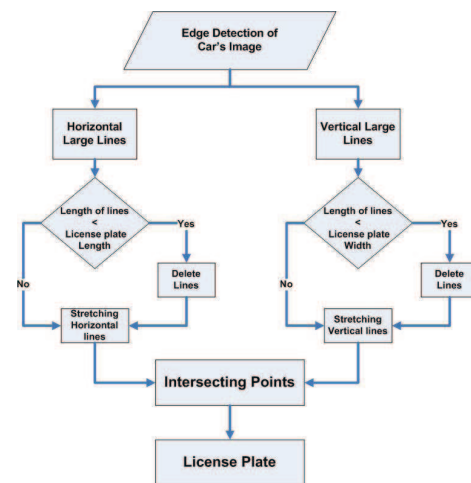


Fig. 1. Algorithm for detection of a License plate in the car's rear image



Fig. 2. (a) Original RGB image with (b) edge image obtained using Sobel operator

Fig. 2(a) shows an input RGB image. Edges detected in Fig. 2(a) are shown in Fig. 2(b). For edge detection Sobel operator has been used. The resultant image is utilized for further steps in detection of the license plate.

In the Fig. 2(b) the background and foreground are represented by logic level "0" and "1" respectively. In order to extract a straight line with no discontinuities, we start reading the pixel values of a row. If a "0" is found, this means that noise has produced discontinuity. Therefore, 8 pixels (or more, depending upon the size of the license plate) are checked below in the vertical direction of the current pixel and if any one pixel below is "1", we replace it with a "0" and the current pixel with "1". The same procedure is applied for all the rows. By doing the above operation we get set of lines which include the desired lines along with the unwanted lines (small lines that do not form a part of the plate) as well as shown in Fig. 3(a).

The same procedure is applied for the pixels in vertical direction. The only difference in this case is that instead of updating pixels in vertical direction (accounting for the line discontinuities), we update the pixels to the right of the candidate pixel as shown in Fig. 3(b).

The next step is to eliminate all those horizontal and vertical lines which are smaller than the approximate width and length of the license plate as illustrated in Fig. 3(c) and in Fig. 3(d), respectively. To get the intersecting points in the actual image, both the extracted horizontal and vertical lines are stretched until where we get intersecting points that correspond to the approximate length and width of the License plate as shown in Fig. 3(e) and Fig. 3(f), respectively.

As it can be observed from Fig. 3(g), there are lot of intersecting points in the image but we compared the distance between the horizontal intersecting points with the defined length of the license plate and mark the points which have approximately the same length as that of the plate. Similarly, the same procedure is applied for the vertical intersecting points as illustrated in Fig. 3(g). This gives the desired intersecting points which are basically the coordinates of the license plate as shown in Fig. 3(h). Once the license plate region is extracted successfully, the next step is to segment individual characters and digits in the region. Discussion of character segmentation is presented in the next section.

III. SEGMENTATION OF CHARACTERS

After license plate region detection, the first step of the segmentation is to threshold the resultant image with fore-

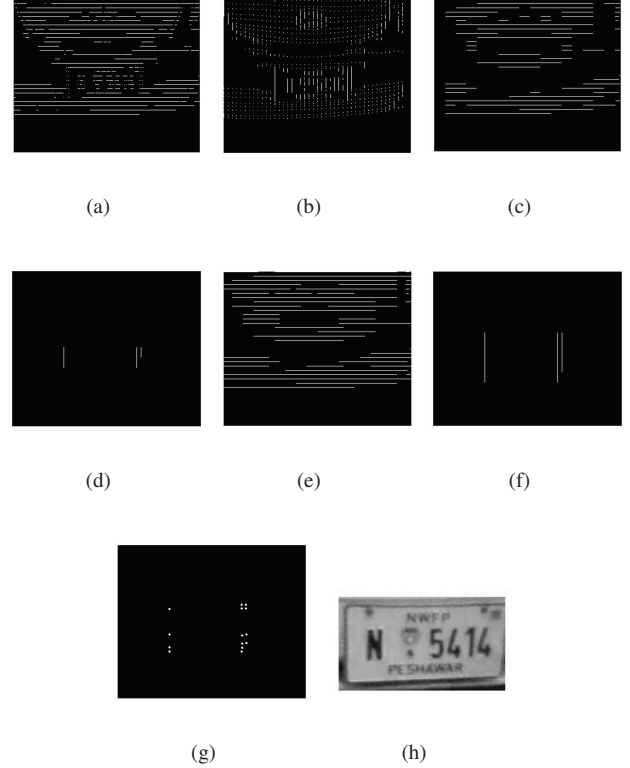


Fig. 3. (a)Horizontal large lines (b) vertical large lines (c) after elimination of horizontal short lines (d) after elimination of vertical short lines (e) stretched horizontal lines (f) stretched vertical lines in order to get intersecting points (g) intersecting points and (h) detected license plate

ground as logical level "0" and background as logical level "1". Thus as shown in Fig. 4(a), it makes the foreground and background of the detected license plate black and white respectively. This process will filter out some unwanted pixels in the detected license plate image and provided a binary image for further cropping of the license plate and extraction of the characters for recognition as illustrated in Fig. 4(d) and Fig. 4(e), respectively. The edge of the binary image is computed again as shown in Fig. 4(b). Subsequently, an array of length equal to the number of rows of the detected license plate is declared and sum of all column pixels for each row of the edge image is stored at corresponding locations in array. This array is processed to extract the portion of binary image between two consecutive minima of the array such that the difference of there indices is maximum as shown in Fig. 4(c).

The resultant image of the above step is shown in Fig. 4(d). After this, an array of size equals to the number of columns of the extracted license plate is declared. All values along the rows for each column of the extracted license plate are added and then stored at the corresponding indices of the array. Portion of the array which starts from zero, ends on zero and having a minimum width much greater than the width of noise pixels are extracted as representation of characters as shown in Fig. 4(e). Similar approach to segmentation is used by [14]. After segmentation of individual characters, it

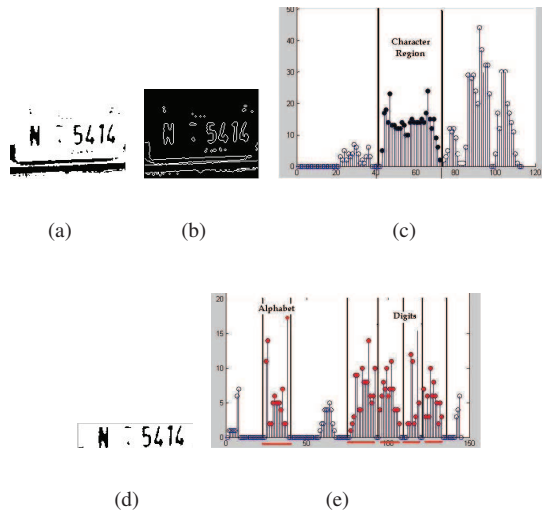


Fig. 4. (a) Binary image of license plate (b) edge of the binary image (c) result of addition of the values of all columns for each row (d) the extracted license plate (e) the result of addition of the values of all rows for each column of the extracted license plate



Fig. 5. The compounds of '0' and '1' are shown in (a) and (b) respectively

is required to recognize them. Recognition of characters and digits is described in the next section.

IV. RECOGNITION OF CHARACTERS

This section is of great importance due to the fact that the discussion includes recognition of segmented characters. Statistical approach for recognition of characters and digits is used because of its ease in implementation on fixed point Digital Signal Processor. For recognition of characters and digits, two types of features are used (a) compound feature, and (b) end points.

A. Compound Features of Characters

A compound is defined as an area that contains pixels with similar values. Each character consists of one or more connected compounds [7]. Since the image is binary, a compound consists of either ones or zeros. The background is only considered as a compound if it appears as a lake within the digit. Fig.5 shows that a '0' has two compounds, while a '1' has only one. On the basis of the compounds, the whole set of characters are divided into three subsets having 1, 2 and 3 compounds.

B. End Point Features of Characters

The number of end points is also a feature that distinguishes digits and characters from each other [7]. It has been observed that the locations of these end points vary, and this can help in distinguishing between the digits. For instance, '6' has only a single end point in the upper half, and none in the lower. The number of end points is found from a skeleton of the character. To obtain the skeleton of a character, a Thinning algorithm is used which results in a one pixel wide skeleton. The end points of the character are the pixels that have only one connected 8-neighbor. The Thinning algorithm [13], [7] iteratively deletes the edges of the character, until a one pixel wide skeleton is reached. The algorithm must fulfill that it does not delete end points and that it does not break connections. Breaking connections will result in additional end points, and therefore this is an important requirement on the algorithm.

C. Algorithm for Recognition of Characters

Firstly, the number of compounds for each digit are found. Number of compound subdivides the whole set of digits into three different categories. After subdivision, digit is recognized on the basis of number of endpoints and their locations. Outline of the algorithm is presented below.

- 1) We have segmented characters.
- 2) First find the compounds.
- 3) Check whether compound is equal to 3.
- 4) If yes, recognize it as a digit "8".
- 5) Else if Check whether compound is equal to 2.
- 6) If yes, check the endpoints, the recognition is now based on the endpoints.
 - Check endpoints whether it is equal to 0, recognize as digit "0".
 - Else If endpoints is equal to 1, check whether the endpoint lies in the upper half portion.
 - If yes, recognized as a digit "6".
 - If not, recognized as a digit "9".
- 7) Else If Check whether compound is equal to 1.
 - If yes, find endpoints.
 - Check whether endpoints lie in right upper corner and left lower corner.
 - If yes, recognized as a digit "5".
 - If not, check whether endpoints are lie in left upper corner and right lower corner.
 - If yes, recognized as a digit "2".
 - If not, check whether both endpoints lie in lower right corner .
 - If yes, recognize as a digit "4".
 - If not, check whether the endpoint lie in the left upper corner.
 - If yes, recognized as a digit "7".
 - If not, recognized as a digit "1".
 - Check whether endpoints are equal to 3.
 - If yes, recognized as a digit "3".

- 8) Character not recognized

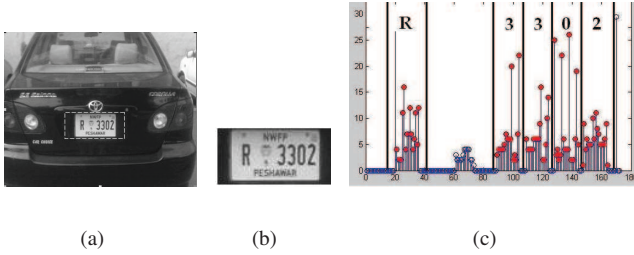


Fig. 6. (a)Test image with a straight camera and good illumination (b) result of extraction of LP region (c) segmentation of the characters

TABLE I
ACCURACY OF OUR PROPOSED LPR SYSTEM

Unit of LPR System	Success Rate	Percentage
Detection of License Plate	51/60	85%
Segmentation	54/60	90%
Recognition of the characters	55/60	91.66%

Similarly, alphabets are recognized by using the same idea of the compounds and endpoints.

V. EXPERIMENTAL RESULTS

Experiments have been performed to test the proposed system and to measure its accuracy. It is implemented on a fixed point Digital Signal Processor and the images for the input to the system are taken as colored images of size of 288x352. Sixty different test input images are taken under various illumination conditions. Fig. 6(a) shows an image that is captured with a straight camera and good illumination. The results on such an image are shown in Fig. 6(b).

License Plates	Illumination	Inclination	Detection	Segmentation	Recognition
	Low	5	Success	Success	Success
	Low	20	Success	Success	Success
	High	0	Success	Success	Success
	Normal	0	Success	Success	Success
	Normal	40	Failure	Success	Failure

Fig. 7. Results of our proposed system

It has been observed that if there is a change in viewing direction i.e greater than 30° , the percentage of failure of our proposed system is more than 50%. Results of different tests are tabulated in Table I.

It has also been observed that accuracy for the detection of license plate region, segmentation of the character and recognition of character is 85%, 90% and 91.66% respectively.

The time taken by our proposed system on the 600MHz fixed point Digital Signal Processor is presented in Table II.

The system is tested on 60 different images. The detection performance can be improved by adding extra condition of

TABLE II
REAL TIME CLOCK READINGS FOR THE BF537

LPR System	Real time clock
Sobel edge	40.8msec
Horizontal and vertical lines	4.9msec
Elimination of horizontal and vertical lines	21msec
Stretching of horizontal and vertical lines	24.15msec
Intersecting points	8.34msec
Cropping of license plate	57.87msec
Detection	157.06msec
Segmentation	6.7msec
Recognition	10msec
Total Time	173.76msec

extracting white region in the detected license plate. In recognition of characters, we can add features such as area, area of the column and area of the row [7].

VI. CONCLUSION

In this paper, an efficient and fast application software is proposed which is designed for the recognition of car license plates. It involves three simple steps of input region extraction, segmentation and then recognition of characters. The proposed algorithm was tested on real images and its been observed that the propose algorithm is providing good results i.e above 85 % accuracy with the limitation of inclination below 30° . In future, this algorithm can be extended to recognize different shapes and colors of car license plates which can have different fonts of characters.

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